

Uncleftish Beholding

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For most of its being, mankind did not know what things are made of, but could only guess. With the growth of **worldken**, we began to learn, and today we have a beholding of stuff and work that watching bears out, both in the **workstead** and in daily life.

The underlying kinds of stuff are the **firststuffs**, which link together in sundry ways to give rise to the rest.

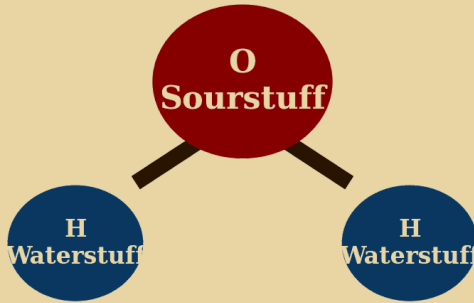
Formerly we knew of ninety-two **firststuffs**, from **waterstuff**, the lightest and barest, to **ymirstuff**, the heaviest. Now we have made more, such as **aegirstuff** and **helstuff**.

The **firststuffs** have their being as motes called **unclefts**. These are mightly small; one seedweight of **waterstuff** holds a tale of them like unto two followed by twenty-two naughts.

Most **unclefts** link together to make what are called **bulkbits**. Thus, the **waterstuff bulkbit** bestands of two **waterstuff unclefts**, the **sourstuff bulkbit** of two **sourstuff unclefts**, and so on. (Some kinds, such as **sunstuff**, keep alone; others, such as iron, cling together in ices when in the fast standing; and there are yet more **ymirstuff yokeways**.)

When unlike clefts link in a **bulkbit**, they make **bindings**. Thus, water is a **binding** of two **waterstuff unclefts** with one **sourstuff uncleft**, while a **bulkbit** of one of the **forestuffs** making up flesh may have a thousand thousand or more **unclefts** of these two **firststuffs** together with **coalstuff** and **chokestuff**.

Water (H₂O) Binding

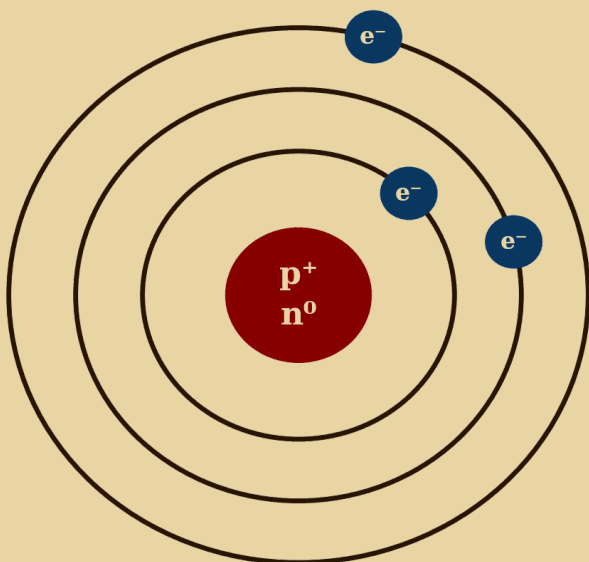


Water (H₂O): a binding of waterstuff and sourstuff

At first it was thought that the **uncleft** was a hard thing that could be split no further; hence the name. Now we know it is made up of lesser motes. There is a heavy **kernel** with a forward **bernstonish** lading, and around it one or more light motes with backward ladings. The least **uncleft** is that of ordinary **waterstuff**. Its **kernel** is a lone **forwardladen** mote called a **firstbit**. Outside it is a **backwardladen** mote called a **bernstonebit**. The **firstbit** has a heaviness about 1840-fold that of the **bernstonebit**. Early **worldken** folk thought **bernstonebits** swing around the **kernel** like the earth around the sun,

but now we understand they are more like waves or clouds.

Structure of the Uncleft



The Uncleft (Atom): kernel and bernstonebits

In all other **unclefts** are found other motes as well, about as heavy as the **firstbit** but with no lading, known as **neitherbits**. We know a kind of **waterstuff** with one **neitherbit** in the **kernel** along with the **firstbit**; another kind has two **neitherbits**. Both kinds are seldom.

The next greatest **firststuff** is **sunstuff**, which has two **firstbits** and two **bernstonebits**. The

everyday sort also has two **neitherbits** in the **kernel**. If there are more or less, the **uncleft** will soon break asunder. More about this later.

The third **firststuff** is **stonestuff**, with three **firstbits**, three **bernstonebits**, and its own share of **neitherbits**. And so it goes, on through such everyday stuffs as **coalstuff** (six **firstbits**) or **iron** (26) to ones more lately found. **Ymirstuff** (92) was the last until men began to make some higher still.

It is the **bernstonebits** that link, and so their tale fastsets how a **firststuff** behaves and what kinds of **bulkbits** it can help make. The **worldken** of this behaving, in all its manifold ways, is called **minglingken**.

Minglingers have found that as the **uncleftish** tale of the **firststuffs** (that is, the tale of **firststuffs** in their **kernels**) waxes, after a while they begin to show ownships not unlike those of others that went before them.

So, for a showdeal, **stonestuff** (3), **glasswortstuff** (11), **potashstuff** (19), **redstuff** (37), and **bluegraystuff** (55) can each link with only one **uncleft** of **waterstuff**, while **coalstuff** (6), **flintstuff** (14), **germanstuff** (22), **tin** (50), and **lead** (82) can each link with four.

Roundaround Board of the Firststuffs

1 H																	2 He						
3 Li	4 Be															5 B	6 C	7 N	8 O	9 F	10 Ne		
11 Na	12 Mg															13 Al	14 Si	15 P	16 S	17 Cl	18 Ar		
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr						
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe						
55 Cs	56 Ba	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu							
87 Fr	88 Ra	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr							
																		113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og

Roundaround Board of the firststuffs

This is readily seen when all are set forth in what is called the **roundaround board** of the **firststuffs**.

When an **uncleft** or a **bulkbit** wins one or more **bernstonebits** above its own, it takes on a backward lading. When it loses one or more, it takes on a forward lading. Such a mote is called a farer, for that the drag between unlike ladings flits it. When **bernstonebits** flit by themselves, it may be as a bolt of lightning, a spark off some **faststanding** chunk, or the everyday flow of **bernstoneness** through wires.

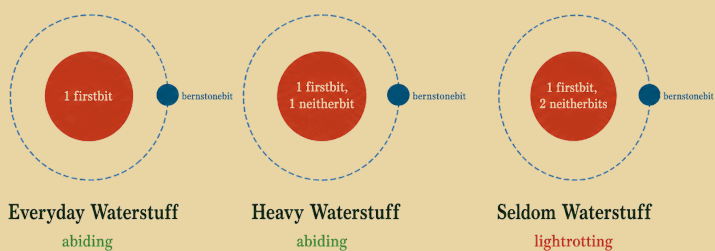
Coming back to the **uncleft** itself, the heavier it is, the more **neitherbits** as well as **firstbits** in its **kernel**. Indeed, soon the tale of **neitherbits** is the greater. **Unclefts** with the same tale of **firstbits** but unlike tales of **neitherbits** are called **samesteads**.

Thus, everyday **sourstuff** has eight **neitherbits** with its eight **firstbits**, but there are also kinds with five, six, seven, nine, ten, and eleven **neitherbits**.

A **samestead** is known by the tale of both **kernel** motes, so that we have **sourstuff**⁻¹³, **sourstuff**⁻¹⁴, and so on, with **sourstuff**⁻¹⁶ being by far the most found.

Having the same number of **bernstonebits**, the **samesteads** of a **firststuff** behave almost alike **minglingly**. They do show some unlikenesses, outstandingly among the heavier ones, and these can be worked to sunder **samesteads** from each other.

Samesteads of Waterstuff



Samesteads (Isotopes) of Waterstuff

Most **samesteads** of every **firststuff** are unabiding. Their **kernels** break up, each at its own speed. This speed is written as the **half-life**, which is how long it takes half of any deal of the **samestead** thus to shift itself. The doing is known as **lighttrotting**. It may happen fast or slowly, and in any of sundry ways, offhanging on the makeup of the **kernel**.

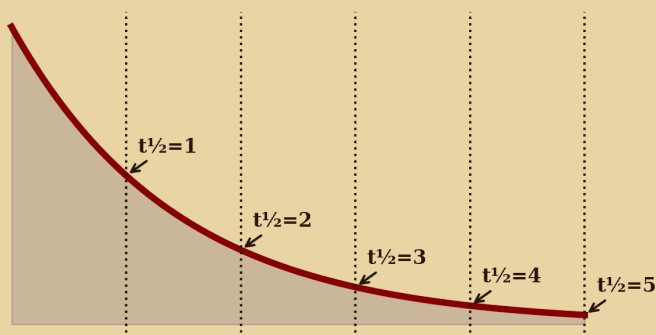
A **kernel** may spit out two **firstbits** with two **neitherbits**, that is, a **sunstuff kernel**, thus

leaping two steads back in the **roundaround board** and four weights back in heaviness. It may give off a **bernstonebit** from a **neitherbit**, which thereby becomes a **firstbit** and thrusts the **uncleft** one stead up in the board while keeping the same weight.

It may give off a **forwardbit**, which is a mote with the same weight as a **bernstonebit** but a forward lading, and thereby spring one stead down in the board while keeping the same weight.

Often, too, a mote is given off with neither lading nor heaviness, called the **weeneitherbit**. In much **lightrotting**, a mote of light with most short wavelength comes out as well.

Lightrotting (Half-Life)



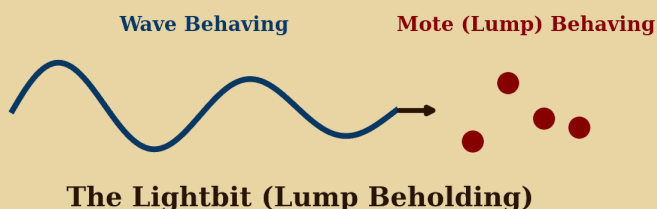
Lightrotting: Samesteads (Isotopes) of Waterstuffial half-life curve

For although light oftenest behaves as a wave, it can be looked on as a mote, the **lightbit**.

We have already said by the way that a mote of stuff can behave not only as a chunk, but as a wave.

Down among the **unclefts**, things do not happen in steady flowings, but in leaps between bestandings that are forbidden.

The knowledge-hunt of this is called **lump beholding**.



Lump Beholding: Waves and Motes

Nor are stuff and work unakin. Rather, they are groundwise the same, and one can be shifted into the other. The kinship is given by the law that work is stuff times the square of the speed of light.

$$E = mc^2$$

E
Work

m
Stuff

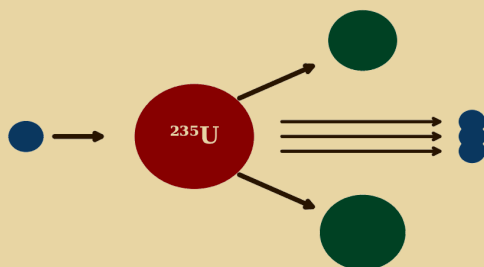
c²
Speed of light

$\mathcal{E} = mc^2$ (stuff and work are groundwise the same).

By shooting motes into **kernels**, worldken folk have shifted **samesteads** of one **firststuff** into **samesteads** of another. Thus did they make **ymirstuff** into **aegirstuff** and **helstuff**, and they have afterward gone on to frame yet heavier kinds that were never known before.

Some of the higher **samesteads** are **splitly**. That is, when a **neitherbit** strikes the **kernel** of one, as for a showdeal **ymirstuff**-235, it bursts into lesser **kernels** and free **neitherbits**; the latter can then split more **ymirstuff** and so bring about a **chainly** doing. **Fastsetly**, this yields a mighty deal of work.

Kernelish Splitting (Fission)

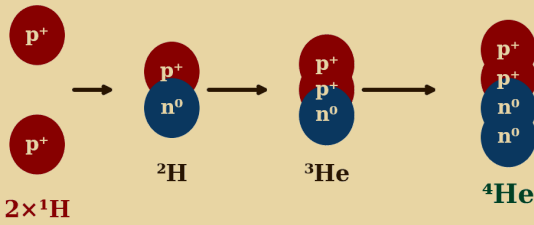


Kernelish Splitting (Nuclear Fission)

With enough strength, lightweight **unclefts** can be made to **togethert**. In the sun, through a row of strikings and **lightrottings**, four **unclefts** of **waterstuff** in this wise become one of **sunstuff**.

Again some stuff is lost as work, and again this is greatly big when set beside the work gotten from a **minglingish** doing such as fire.

Togethermelt (Fusion)



Togethermelt (Nuclear Fusion) in the Sun

Today we wield both kind of **uncleftish** doings in weapons, and **kernelish** splitting gives us heat and **bernstoneness**.

We hope to do likewise with **togethermelting**, which would yield an unhemmed wellspring of work for mankindish goodgain.

Soothly we live in mighty years!